The Islamia University of Bahawalpur

Department of Information technology

SOFTWARE REQUIREMENTS SPECIFICATION
(SRS DOCUMENT)

for

AI Based Resume Screening System with Web Interface

By

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Revision History

Name	Date	Reason for changes	Version

Application Evaluation History

Comments (by committee) ***include the ones given at scope time both in doc and presentation**	**Action Taken**

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Signature_____

1 Introduction

Recruitment and hiring is a critical process in every organization, yet manually screening resumes (CVs) is **time-consuming and prone to human bias**. With the increasing number of applicants, HR teams often struggle to shortlist the best candidates efficiently.

Solution: The proposed project is an **AI Based Resume Screening System** which automatically extracts candidate details from resumes, evaluates their skills, education, and experience, and ranks them according to the requirements of the job role. The system uses **AI techniques including NLP (Natural Language Processing) and ML (Machine Learning)** to ensure accurate evaluation. A **web interface** provides ease of use for both administrators and candidates.

1.1 Purpose

The purpose of this document is to define the functional and non-functional requirements of the **AI Based Resume Screening System with Web Interface**. This system is designed to assist Human Resource (HR) departments in automating the process of resume evaluation and candidate shortlisting.

The traditional resume screening process is time-consuming and often influenced by human bias. This project aims to solve these issues by using **Artificial Intelligence (AI), Natural Language Processing (NLP), and Machine Learning (ML)** techniques to analyze resumes and rank candidates based on job requirements.

1.2 Scope

The proposed system is a web-based application that automates the resume screening process. It allows candidates to upload their resumes and enables HR personnel to view ranked candidates based on AI analysis.

The system will include the following functionalities:

- Resume upload and storage
- Extraction of key information such as skills, education, and experience
- AI-based analysis and candidate scoring
- Ranking of candidates according to job requirements
- Admin dashboard for managing job roles and viewing results
- Secure database for storing all records

In future enhancements, the system can be extended to support job recommendation systems and integration with online job portals.

2 Overall description

2.1 Product perspective

The system is a standalone web application designed to automate and improve the recruitment process. It follows a **three-tier architecture**:

- **Frontend:** User interface developed using HTML, CSS, and JavaScript
- **Backend:** Python Flask handles application logic and AI processing
- **Database:** SQLite/MySQL used for storing user data, resumes, and results

The system acts as an intelligent assistant for HR departments by automating resume analysis and ranking.

2.2 Operating environment

The system will operate in the following environment:

- Operating Systems: Windows, Linux, macOS
- Programming Language: Python 3.9
- Framework: Flask (Web Framework)
- Database: SQLite or MySQL
- Web Browsers: Google Chrome, Mozilla Firefox, Microsoft Edge

Required libraries include:

- NLP: NLTK, spaCy, TextBlob
- ML: Scikit-learn
- File Processing: pdfminer.six, python-docx

Since the system is web-based, it can run on any device with a modern web browser.

2.3 Design and implementation constraints

The development of the system is subject to the following constraints:

- The system will be developed using open-source and free tools
- Resume formats are expected to be in standard PDF or DOCX format
- Initial implementation will use limited datasets
- Internet connection may be required for deployment or optional AI services
- The system is designed as an academic project and not optimized for large-scale production use

3 Requirement identifying technique

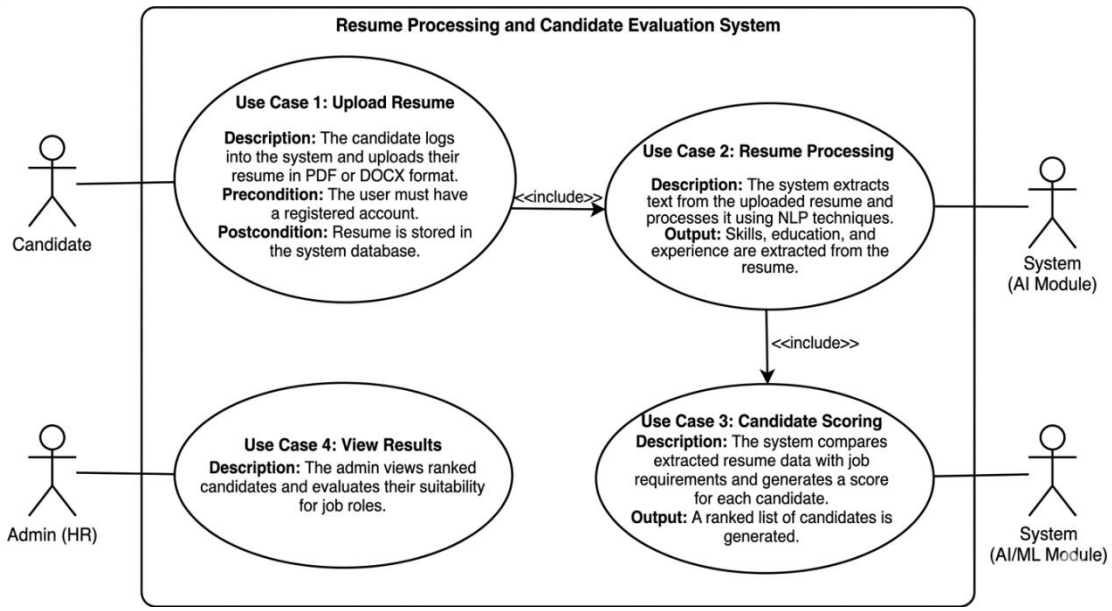
To define the functional and non-functional requirements for this project, the following techniques were employed:

The requirements of the system have been identified using the following methods:

- Analysis of existing resume screening systems
- Study of recruitment processes in HR departments
- Literature review of AI and NLP applications in recruitment
- Use case modeling to understand system behavior

3.1 Use case diagram

Use case diagram using **MS Visio** for system. For detail guideline to developed use case diagram



3.2 Use case description

The system is divided into two main layers: the high-level Learning Community Detection System (User Interface layer) and the internal Community Detection Pipeline (Processing

layer).

Table 5 Show the detail use case template

Use Case ID:	Enter a unique numeric identifier for the Use Case. like. UC-1
UC-01	Upload Resume
Primary Actor	Candidate
Description:	The candidate logs into the system and uploads their resume in PDF or DOCX format.

UC-02:	Resume Processing
Primary Actor	System (AI Module).
Description:	The system extracts text from the uploaded resume and processes it using NLP techniques.
Key Tasks:	Analyze the resume and extract key things from resume.
Outcome:	Skills, education, and experience are extracted from the resume.

UC-03:	Candidate Scoring
Primary Actor:	System (AI/ML Module)
Description:	The system compares extracted resume data with job requirements and generates a score for each candidate.
Outcome:	A ranked list of candidates is generated.

UC-04:	View Results
Primary Actor:	Admin (HR)
Description:	The system compares extracted resume data with job requirements and generates a score for each candidate.
Outcome:	A ranked list of candidates is generated.

4 Functional Requirements

This section describes the functional requirements of the system expressed in natural language style. This section is typically organized by feature as system feature name and specific functional requirements associated with this feature. It is just one possible way to arrange them. Other organizational options include arranging functional requirements by use case, process flow, mode of operation, user class, stimulus, and response depend what kind of technique which has been used to understand functional requirements. Hierarchical combinations of these elements are also possible, such as use cases within user classes.

4.1 Functional Requirement

The system shall provide the following functionalities:

1. The system shall allow users to register and log in securely.
2. Candidates shall be able to upload resumes in PDF or DOCX format.
3. The system shall extract text from uploaded resumes using NLP techniques.
4. The system shall identify key information such as skills, education, and experience.
5. The system shall evaluate and score candidates based on job requirements.
6. The system shall generate a ranked list of candidates.
7. Admin users shall be able to manage job roles (add, edit, delete).
8. Admin users shall be able to view candidate rankings and reports.
9. All data shall be stored securely in a database.

Table Show the Functional Requirement template

Requirement ID	Requirement	Source	Rationale	Dependencies	Priority
FR-01	The system shall allow users to register and log in securely.	Stakeholders / HR System	To ensure only authorized users access the system.	Database, Authentication Module	High
FR-02	The system shall allow candidates to upload resumes in PDF or DOCX format.	User Requirement	To collect candidate data for analysis.	File Upload Module, Storage System	High
FR-03	The system shall extract text from uploaded resumes using NLP techniques.	System Design Requirement	To convert unstructured resume data into usable text format.	Resume Upload Module, NLP Library	High
FR-04	The system shall identify key information such as skills, education, and experience.	HR Requirements	To evaluate candidate qualifications automatically.	Text Extraction Module	High
FR-05	The system shall evaluate and score candidates based on job requirements.	HR Decision Process	To rank candidates objectively using AI algorithms.	Feature Extraction, ML Model	High
FR-06	The system shall generate a ranked list of candidates.	HR Workflow	To help recruiters shortlist top candidates easily.	Scoring Module	High
FR-07	The system shall allow admin users to manage job roles (add, edit, delete).	Admin Requirements	To maintain updated job postings and requirements.	Admin Panel, Database	Medium
FR-08	The system shall allow admin users to view candidate rankings and reports.	HR Requirement	To support decision-making during recruitment.	Scoring Module, Database	High
FR-09	The system shall securely store all user data, resumes, and results in a database.	System Requirement	To ensure data persistence and security.	Database System	High
FR-10	The system shall ensure only authenticated users can access sensitive data.	Security Requirement	To protect candidate and HR information.	Authentication System	High

5 Non-Functional Requirements

This section specifies nonfunctional requirements other than functionality” presents more information about these quality attribute requirements and many examples.

5.1 Algorithmic Efficiency in Large-Scale Environments

The system is designed to process resumes efficiently using optimized NLP and ML techniques. Although it is an academic project, the following considerations are taken:

- Resume parsing should be performed in minimal time
- Feature extraction must be optimized for fast processing
- The system should be capable of handling multiple resumes efficiently
- The architecture should support future scalability

5.2 Performance

The system is expected to meet the following performance criteria:

- Resume processing time should remain under 3–5 seconds per file
- Web interface should respond within 2 seconds for user actions
- Database queries should be optimized for fast retrieval
- The system should support multiple concurrent users without performance degradation

6. References

In a Software Requirements Specification (SRS), the **Reference** section lists all the materials, standards, and documents used to define the requirements for the **Learning Community Detection System**.

6.1 Project Documentation

The following documents and academic materials were consulted during the preparation of this project:

- IEEE Software Engineering Standards for Software Requirements Specification (SRS)
- University Final Year Project Guidelines and Templates
- Existing Final Year Project reports related to HR systems and resume screening systems
- Academic lecture notes on Software Engineering and System Analysis & Design
- Industry case studies related to recruitment automation systems

These documents helped in understanding the structure, format, and required components of a professional software engineering project.

6.2 Technical and Algorithmic Sources

The following technical resources, libraries, and frameworks were used or studied for implementation purposes:

- Python Official Documentation – https://docs.python.org
- Flask Web Framework Documentation – https://flask.palletsprojects.com
- Scikit-learn Machine Learning Library – https://scikit-learn.org
- Natural Language Toolkit (NLTK) Documentation – https://www.nltk.org
- spaCy NLP Library – https://spacy.io
- TextBlob Library for Sentiment and Text Processing – https://textblob.readthedocs.io
- pdfminer.six for PDF text extraction
- python-docx for DOCX file handling
- Research papers on “Resume Screening using Machine Learning and NLP”
- Kaggle datasets for resume and job matching analysis

These sources were used to understand and implement the AI-based resume parsing, feature extraction, and candidate ranking mechanisms.